EE5311 CA2 Assignment (AY25/26 Semester 2)

This assignment counts toward 25% assessment for the course.

This assignment is due for submission by **8 March 2026 (11:00 pm)**.

The assignment problem below is open-ended and may admit many possible solutions. Bearing in mind that the assignment provides you an opportunity to practice what you learn in this course, it is recommended that you think about how physics-guided machine learning techniques might help you solve the problems.

You **may** discuss the assignment problem and solution methodologies with other students and the instructor, but you are **not allowed** to share code or project report with other students. The Canvas discussion forum is a great place for discussions and/or clarification regarding this assignment.

Any copying or plagiarism will hamper your own learning, and may also result in disciplinary action against you.

Problem statement

You work for a marine survey company that has recently laid an undersea acoustic sensor array (underwater microphones) on the seabed. You have been tasked with estimating the shape of the array, i.e., the location of each sensor on the seabed. The array consists of 1926 sensors connected in series with 1 m cables. Each sensor is 2.13 cm in size, and thus the spacing between sensors cannot exceed 1.0213 m.

The array was laid from a ship. The track of the ship was estimated using a GPS logger and is provided to you in the file `ca2-track.csv`. The positions in the file are in meters. While the array was laid from the stern of the ship as it moved along the track, its location on the seabed may somewhat deviate from the track due to prevailing currents pushing the array as it was laid. You may assume that the seabed in the area where the cable is laid is flat with a water depth of 20 m.

Several acoustic transmissions were made after laying the cable to aid with your estimation. The ship moved to 6 different locations (given in file `ca2-transmissions.csv` in meters) and deployed a transmitter at a depth of 2 m. From each location, 2 acoustic transmissions were made in quick succession. The transmissions were received at many of the acoustic sensors in the array. Noisy estimates of time of reception of the acoustic signal at each of the sensors are available in the file `ca2-timings.csv` (in seconds). The estimated reception times are given with respect to arbitrary time origins (different for each transmission), and the time at which each transmission was made is unknown. The sensors are all synchronized and therefore the time difference of arrival between sensors provides information about relative positions of the sensors. Columns `t11` and `t12` contain estimates of time of arrival of the first and second transmissions from transmitter location 1 at each sensor. Similarly columns `t21` and `t22` refer to the estimated times of arrival of signals from transmitter location 2 at each sensor. Columns `t31`, `t32`, `t41`, `t42`, `t51`, `t52`, `t61` and `t62` refer to the estimated times of arrival of pairs of signals from transmitters 3, 4, 5 and 6 respectively. Do bear in mind that the estimates are noisy, and sensors where the signals are poorly heard may have very significant timing errors.

You may assume that the speed of sound in water is 1540 m/s.

Submission requirements

(25 marks)

You are required to submit a `zip` file (name the file with your login ID, e.g., `e1234567.zip`) containing the following:

- Brief report (**page limit: 2 pages**) in `pdf` format (please do NOT submit Word documents!), clearly outlining the method used and the rationale for your design choices.
- A CSV file named `results.csv` containing your results. The file should have 3 columns (`sensor`, `x` and `y`), with the first column being sensor number (1 to 1926), and the next two columns being estimated x and y positions of the sensors (in meters).
- Well-commented runnable code (in the language of your choice) yielding your results. Running this code should create an identical `results.csv` file as your submission. You should include a `README.txt` file with instructions on how to run your code.

All filenames and format of the `results.csv` must be exactly as described above. If the file format is different, the file may not be considered for grading.